



# Role of hub and spoke model for ICTs in agriculture

Nancy J. Anabel<sup>1</sup> · Priyanka Mohan<sup>2</sup> · R. Rajkumar<sup>1</sup>

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**Abstract** Small-holding farmers are in constant need of dynamic, timely, scientific information and knowledge on good agronomic practices, government entitlements, and market information, including access to early warnings, and application of new technologies, to better enhance their agricultural production. However, the inadequate proportion of agri-extension workers to farmers has left a void in providing timely agro advisories and inputs to support farming interventions. This has hindered farmers' ability to take informed decisions for bettering their yields and income. The role of information and communication technologies (ICTs) and its potential to bridge the knowledge, digital, and gender gaps in agriculture remained unexplored and such technologies were accessible only to the urban rich. In the 1990s, M. S. Swaminathan Research Foundation, through an interdisciplinary dialogue on 'Reaching the Unreached: Information Technology' evolved the concept of using ICTs to link small farmers to need-based, demand-driven information. This information village transpired as a hub and spoke model of Village Resource and Knowledge Centres connecting the knowledge seekers (farmers) with knowledge holders (researchers, government institutions, and agriculture-based Non Governmental Organizations and progressive farmers) through land–lab, lab–land, land–land, lab–land

communication. This paper reviews the evolution of this model, its results on the ground and contemporary relevance.

**Keywords** Information · Hub and spoke · Village Resource Centre · Village Knowledge Centre · Knowledge · ICT · Gender · Agriculture · Extension

## 1 Introduction

Small-holding farmers are in constant need of timely real-time scientific information and knowledge to better their agricultural production. Information on good agronomic practices, government entitlements, and market information, including access to early warnings, and application of new technologies are of importance to support them during their agricultural operation. The agriculture extension system has often left the smallholding men and women farmers untouched in addressing their needs [1]. The Training and Visit (T & V) system of extension, funded by World Bank improved the intensity of extension in the 1980s [2] and the decline in the external support for T & V system in the early 1990s left the states in India to manage the extension system on their own, which had the adverse effect of weakening the public sector extension system [3]. "By late 1970 and 1980s, the agricultural extension service in India was becoming progressively weak and even outdated" and in the early 1990s ICT were beginning to impact the urban rich which were yet to be accessed by the rural communities [4]. At the same time, the all India extension effectiveness index, ranked the extension mechanism in India as moderately effective [5].

The existing ratio of only one extension worker for every 2879 farmers in India [6] signifies an inadequate link

✉ Nancy J. Anabel  
anamssrf@gmail.com

<sup>1</sup> Information, Communication and Education, M. S. Swaminathan Research Foundation, Taramani, Chennai, India

<sup>2</sup> Development Consultant, Information, Communication and Education, M. S. Swaminathan Research Foundation, Taramani, Chennai, India

between research, extension, and farmers [7] leading to a wide gap between the proportions of technologies developed by the public agricultural research system and technologies adopted by the farmers [8]. A study on understanding the needs of community in South of Tamil Nadu and Puducherry by M. S. Swaminathan Research Foundation (MSSRF) in 1991 also concluded the need for disseminating appropriate knowledge and building skills on latest technologies for the farming community [9].

The diversified socio-economic-cultural aspects of the farming community demand contextual and season-specific agricultural extension system that enables them to overcome problems in crop cultivation. Overall, poor access to information and crumbling extension services resulted in low productivity and slow uptake of information [10].

In order to better equip farmers to cope with the current and evolving challenges, M. S. Swaminathan Research Foundation (MSSRF) had brought together linkages between the traditional approach of land to land (farmer to farmer) communication, enabling peer learning among farmers and lab to land (scientist to farmers) communication to enable farmers to interact with the knowledge holders directly, to better equip them to manage the challenges in agriculture [9]. In the process of applying these specific links, MSSRF envisioned the important role that ICT can play in agriculture and sustainable rural development, as an accelerator, for facilitating timely delivery of value-added information to enable farmers to undertake informed decisions.

A user-centric system that disseminates value-added information for knowledge empowerment was designed by MSSRF in 1992. The idea of this 'information village' was to meet the needs of farmers using advanced technology [9]. MSSRF piloted ICTs by using the hub and spoke model to establish Village Resource Centres (VRC) and Village Knowledge Centre (VKC) in 1997. This ICT-based model improved the availability of timely information as knowledge to the farming community by bridging the knowledge, digital, and gender gaps. This paper reviews the evolution of the hub and spoke model, its contemporary relevance, and benefits to the small and marginal farming community.

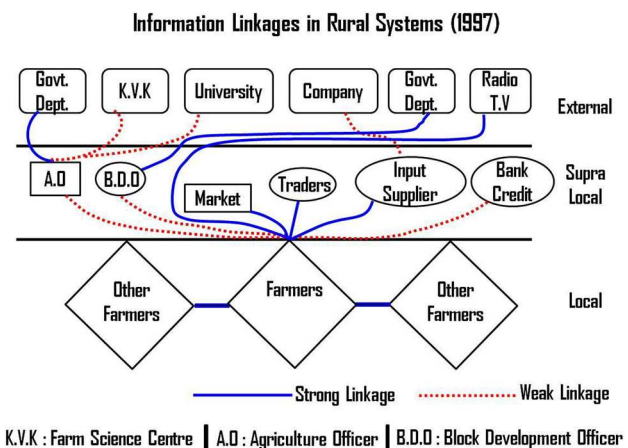
## 2 Hub and spoke model of village resource and knowledge centres

The strong disconnect between farmers and knowledge holders underpin the need for timely information and knowledge services. The potential of ICT to bridge such a knowledge gap led MSSRF to evolve the concept of information villages in 1992.

### 2.1 Information and knowledge gap among farmers

A wide range of information services are required at different stages of farming. These services include production technology, access to quality inputs, government entitlements and markets, credit linkages, infrastructure and business development services among farmers. The situation analysis conducted by MSSRF in Puducherry covering 10% of the resident families showed the dependency of farmers on informal sources (local shopkeepers, the market place, and the suppliers) for information. The study clearly brought out the existing information linkages in the rural system at three levels namely local, supra local, and external as depicted in Fig. 1. The bottom layer presented in Fig. 1 indicates that there is a strong linkage between farmers to source information. Similarly, the farmers' link with markets, traders and input suppliers at the middle level connotes a strong linkage in terms of information flow. The study brought out the strong information flow to farmers through the broadcasting units such as radio and television, at the external level. However, in contrast, the study highlighted the weak flow of information between the farmers and agricultural officer, block development officer, and credit institutions. It became apparent that the agriculture officers have a poor linkage with the Krishi Vigyan Kendra (KVK) and Universities, which are the primary sources for equipping farmers with knowledge, skills, and technology choices.

The study findings are supported by the 2005 situation assessment survey on Farmers Access to Modern Technology for Farming of the National Sample Survey Organization (NSSO) which notes that 60% of farmers did not have access to any source of information on modern technology to support their farming practices. Those farmers who received any information noted that their main sources of information are progressive farmers and input



**Fig. 1** A study result of information linkages in rural system at Puducherry, 1997 by MSSRF



VKC in a rent-free building that is accessible to all sections of the community, with trained Knowledge Workers (KWs), a man and a woman, to manage the centre and disseminate information and knowledge to the farmers, fishers, landless labourers, men, women, and children. The experiment on the other hand strengthened the model by its bottom up approach, community ownership, and inclusive development by enabling access to information, localizing the networked information into a usable knowledge by the poor.

### 2.3 Strengthening of hub and spoke model

In 2003, the VRC and VKC programmes were further strengthened with the creation of the Jamsetji Tata National Virtual Academy to collaborate with several international and national partners for developing content and capacity building for sustainable rural development. This concept was conceived through a three-tier system, demonstrated using hub and spoke model—the national hub, block hub, and VKCs. In 2003, the learning from the testing of hub and spoke model was used to strengthen and expand the model across Tamil Nadu, to facilitate timely delivery of need-based knowledge services to the farming community. In the refined model given in Fig. 3, in addition to the block-level VRC, a national-level hub was introduced with the provision of uplink and downlink satellite connection in partnership with the Indian Space Research Organization (ISRO).

The key components of the model, such as Content (locale-specific, demand-driven, relevant, dynamic, and static), Connectivity, Capacity Building (of grassroot torchbearers), Convergence (of knowledge stakeholders from public, private, civil society, and community centric) and Care and Management (of knowledge centres), with their interconnectedness played a significant role in ensuring that the products and processes are in place. Content and capacity building have provided the community with the required knowledge and skills. The Connectivity component involved a digital platform as a means to reach out to the community at large so they can undertake informed decision-making. Convergence helped leverage and establish strategic partnerships with diverse and appropriate knowledge holders, and in turn enabled the VRCs to value add it as an actionable knowledge and provide knowledge to the community according to their needs. Care and management guarantees the process of community ownership leading to sustainability and result orientation of the model.

Convergence established the essence of participatory science communication between lab-to-land, land-to-land<sup>4</sup>

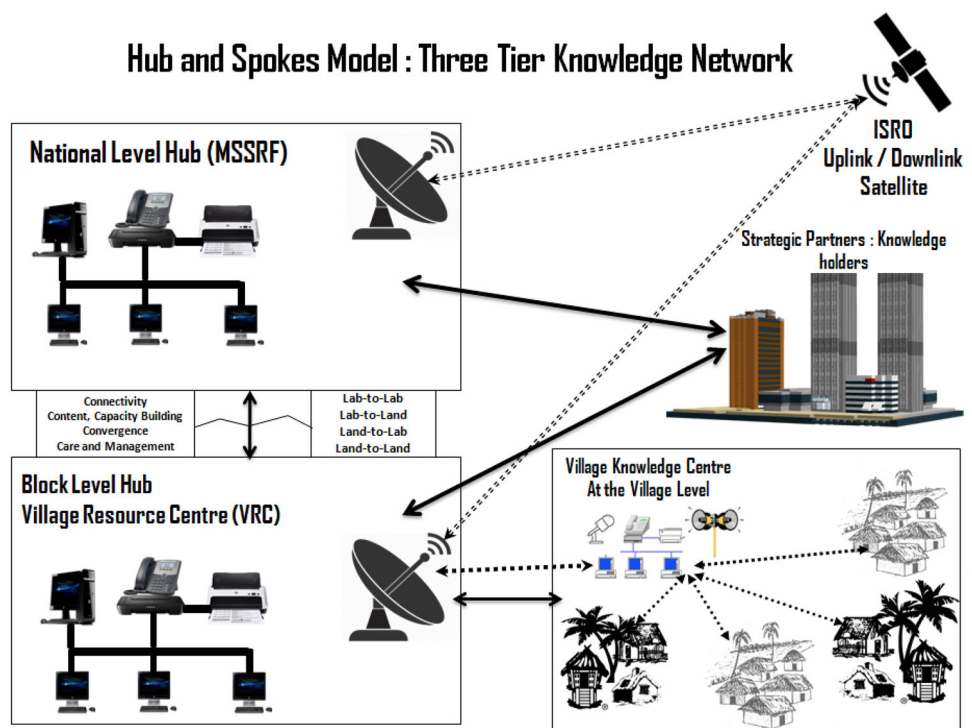
and land-to-lab [14] to break existing innumerable silos by bridging the existing knowledge, gender, and digital divides. The farmers who are endowed with pragmatic experience and traditional knowledge on farming are identified in different villages. Knowledge-sharing between these farmers and those in other locations who do not have such knowledge was facilitated using video and audio conferences. This has ensured an information network among farmers to reach out to their known hubs to address their information needs. For example, Kalaivani Rajendran, a woman farmer in Thanjavur district was the first to introduce System of Rice Intensification (SRI) of rice production and usage of drum seeder for direct sowing to fellow farmers. This has been observed by farmers of her village and neighbouring villages from different blocks in Thanjavur district. Such type of farmer to farmer learning helped them improve production with less input. The scientists were connected through videoconferences periodically to help farmers understand new practices and techniques in farming. For example, a virtual course for 3 months using video conference was done which connected experts from ICRISAT with farmers to guide them on the techniques of groundnut cultivation. This helped them access best practices and techniques, season specific advisories, and practical recommendations to cope with and better manage field level challenges in farming. Specific interactions were organized between the scientists and farmers to understand existing issues in farming, besides providing feedback on the information shared. Another study on Human dimensions of ICT in rural India—Reflections by rural women and men conducted by MSSRF in 2009 emphasized that the horticulture and agriculture department officials use the VKCs as a contact point to meet the villagers. For example, in the VKC at Merpanaikadu of Pudukottai district, Malarvizhi, a young widow, was inspired to raise a kitchen garden after attending a horticultural discussion conducted by the VKC with the help of the Department of Horticulture and the Horticultural College. She said, “I am a poor widow and I own a small piece of land (1/2 acre). The discussion and demonstration of different methods of cultivating plants and crops by the horticulture department officials through VKC was helpful. They also distributed good quality vegetable seeds and saplings to plant in the vacant plot near my house. We had many varieties like bottle gourd, ash gourd, snow peas, ladiesfinger, brinjal, drumstick, tomato, and curry leaves. After sowing and planting we maintained them well and now I am able to use the vegetables for my household consumption and also sell to my neighbours. I found this profitable” [15].

All these elements became key features that enhanced the holistic nature of the hub and spoke model, making it an open access system addressing the information and

<sup>4</sup> Lab denotes scientists and land denotes farmers.



**Fig. 3** A three tier structure of hub and spoke model demonstrated in 2003



knowledge needs of the farming community through a pro-women, pro-nature and pro-poor lens.

## 2.4 Evolution of different models

The demonstration of hub and spoke model has led many partners to evolve different models of ICTs in the early 2000, such as the *e-choupal*, *e-seva*, Indian Farmers Fertiliser Cooperative Limited (IFFCO) IFFCO Kisan Sanchar Limited (IKSL), Reuters Market Light (RML), Warana, Ekgoan, and Media Lab Asia for reaching communities across India who are not connected to such networks and to integrate e-agricultural services. The hub and spoke model underpinned the formulation of “Mission 2007: Every Village a Knowledge Centre”, through an alliance of stakeholders across India to encourage different stakeholders to promote knowledge empowerment among communities using different models. This alliance was later named Rural Knowledge Movement (*Grameen Gyan Abhiyan*).

## 2.5 Evolving knowledge hubs: VRC, VKC and ICT action platforms

With the changing trends in technology, the once popular and in demand ICTs can become redundant, giving way to new technologies. However, the data shows that the model of VRC and VKC continues its role and remains relevant in today’s context as it caters to the dynamic information

needs of both men and women farmers in the context of climate change and its uncertainties. This is possible due to its openness to accommodate relevant technologies that suit the rural context and community.

The hub and spoke model has now become a home for a variety of ICTs to function either as a subset of the model or as an exclusive spoke in itself to disseminate knowledge to the community. The ICT action platform encompasses a combination of existing and upcoming tools including audio conferences, mobile applications, mobile-based helpline, phone-in dialogue, videoconference, social media (Facebook, WhatsApp, YouTube etc.), internet, Massive Open Online Courses (MOOC), artificial intelligence, and webinars.

The hub and spoke model has allowed the seamless inclusion and testing of a gamut of technologies since 1998. Table 1 clearly indicates the chronological order of ICTs used over a period of time based on the need. These are synchronized in pace with the changing trend in technology to fulfil the knowledge needs of the community besides strengthening and nurturing their ability to handle technology and access information.

## 3 Evaluation

A systematic study was conducted using primary and secondary data to understand the evolution of hub and spoke model and its impact among the farming community.

**Table 1** Chronological order of ICTs used over a period in the hub and spoke model

| Year      | Tools used for knowledge dissemination                                                                                                                                                                                                                                                                                     | Technologies/applications                                             |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1998–2001 | Motorola VHF for data\voice transfer and functions as walkie-talkie—VHF technology line speed is below 1024 kbps                                                                                                                                                                                                           | Very high frequency technology                                        |
| 2001–2003 | Two way communication using 853 Trunk Controller—VHF technology speed is below 1024 kbps                                                                                                                                                                                                                                   |                                                                       |
| 2003–2005 | Two way communication using Spread Spectrum technology with 11 Mbps Speed (RF), frequency range 2.4 GHz (802.11b)                                                                                                                                                                                                          | Radio frequency technology                                            |
| 2004–2007 | Wireless Public Address System (W-PAS) frequency range 2.4 GHz (802.11b). PAS—spread spectrum technology                                                                                                                                                                                                                   |                                                                       |
| 2002–2005 | Community radio—Open Knowledge Network (OKN)—Satellite Radio, Satellite based uplink and downlink satellite facilities of Indian Space Research Organization                                                                                                                                                               | Satellite communication                                               |
| 2006–2008 | Fisher Friend Mobile Application (FFMA) in CDMA platform. Broad band connection, SMS, voice message services                                                                                                                                                                                                               | CDMA technology                                                       |
| 2009–2012 | GSM Based (DTMF controller) Public Address System (PAS), phone in program tool (portable PAS with audio controller)<br>Village knowledge portal using open source software, Drupal                                                                                                                                         | GSM technology                                                        |
| 2009–2018 | Internet based video conferencing (ooVoo, Skype, Go to meeting, etc.)                                                                                                                                                                                                                                                      | DSL (Digital Subscriber Line)                                         |
| 2011–2013 | Mobile plus—3G video conferencing, GSM based (Arduino Controller) PAS, FFMA, e-plant clinics, Web based Knowledge Management System (KMS)                                                                                                                                                                                  | GSM technology, Android                                               |
| 2014–2018 | e-plant clinics, In-house—voice blast services, audio and video conferencing, bridge and Interactive Voice Responses (IVR) solution, MOODLE based capacity building, pan India Fisher Friend Mobile Application, mobile applications for agriculture, WhatsApp, Facebook, Webinars, YouTube, Twitter, video based learning | Fibre optical, Android, MOOC, MOODLE<br>Social media, Graphical tools |

A variety of secondary information sources including reports, research articles, and anecdotes were reviewed to gather data to gain insights on the significant results of the hub and spoke model, and its applicability in ICT and agriculture.

To understand the significant contribution of the VRC and VKC model using ICTs among farmers in agriculture, both quantitative and qualitative data were collected from the primary stakeholders. A simple random sampling method was used to identify 300 farmers (182 men and 118 women) who are the users of VRC and VKC in Pudukkottai (2004) and Thanjavur (2004) districts in Tamil Nadu State for the period of April 2017 to March 2018. A structured open-ended questionnaire was used for primary data collection from the respondents to assess the role played by the hub and spoke model and its results among the farming community. The qualitative data were collected in the form of case studies.

## 4 Results and discussion

This section presents results in two parts. The first part focuses on the results brought out by different stakeholders through studies. The second part of the results is brought out through the study carried out by MSSRF.

### 4.1 Results of hub and spoke model as observed by others

The impact of the hub and spoke model starting from its genesis to the current phase is well-documented by several research and evaluation studies, experts, reports, and print media. This section presents a few of those results as studied by diverse stakeholders:

#### 4.1.1 Inclusive approach for knowledge empowerment

The MSSRF “revolutionized and pioneered” the utility of ICTs for development in agriculture through Hub and Spoke model of VRC and VKC among rural communities for providing access to locale-specific, demand-driven information using modern technology [16]. The inclusive approach emphasized through the measures of social stratification in the community helped ensure that the entire community benefitted from the VKC regardless of caste, creed, gender, and class [17]. The participatory approach engaged village community and stakeholders consistently and continuously to regenerate the idea and seek affirmation among the community. The strong sense of ownership nurtured and demonstrated among the village communities is an important feature of the VKC model [18].

“In many cases, the telecentre (VKC) is intentionally opened in areas inhabited by disadvantaged sections, forcing residents from the ‘upper strata’ of society to come to these places. This does help change the power equations, albeit slowly. Inclusion of disadvantaged communities is a pre-requisite for the opening of a telecentre in any area. This initiative is one of those rare examples that successfully combine the issues of financing, community monitoring and ownership. This initiative thrives and will continue to do so because it caters to the information needs of the local people, gives them a sense of ownership of the initiative, and the chance to shape its running. This makes the centre indispensable to the lives of the community” [19].

#### 4.1.2 *Information network and women empowerment*

“Bridging the digital divide is a powerful method of bridging the gender divide” [20]. The uniqueness of the information village has attracted global attention and there have been detailed reports of the women-managed Knowledge Centres in the New York Times and the International Herald Tribune [9]. “The women in particular have gained many intangible benefits in addition to enhanced opportunities for rural livelihoods.” [4]. “In Embalam, especially women associated with the VKC, have seen tremendous improvements in their self-confidence and self-esteem, which has enabled them to improve their and others incomes in their village. For the socially excluded status of women, VKC’s provided transformative remedy, whereby the role of women as perceived by Embalam’s community underwent a radical change over time. These women are now seen as sources of reliable information as is the VKC” [21].

“The MSSRF Information Village programme increases the status and influence of women by making them the primary local knowledge providers. Drawing on this concept, I envision a global electronic network that connects scientists to people at all levels - farmers’ organizations and village women. The network will allow them to easily access the scientific and technical knowledge that they need to solve local problems and enhance the quality of their lives, as well as to communicate their own insights and needs back to scientists” [22].

#### 4.1.3 *Locale-specific knowledge through ICTs and partnership*

Knowledge connectivity confers multiple economic and social benefits. The ICT helps to improve the timeliness and efficiency of farm operations and enhances income through producer oriented markets [20]. The hub and spoke model clearly defined information hierarchy to ensure that

functioning and sharing of information is easy and effective [23] and its role in creation of and updating relevant content to suit to the local needs is a key element of this model [3]. As a result, the VKCs have shown improvements in livelihoods of the rural community, by providing timely information and knowledge on agriculture, animal husbandry, fisheries, health, education and Government schemes [24].

The hub and spoke model became an open-access system with pillars of partnership hence revolutionizing it as a potential sustainable model in rural development for enhancing livelihoods of farming community. It is a notable cutting edge model of ICTs that drew attention as a means to deliver rapid knowledge services to its farmers, reduce the divide between the rich and poor, and urban and rural. “The diffusion also depended on the relevance, efficiency, policy support and linkages to markets” [25]. MSSRF was able to deploy appropriate technology, based on the results from the needs assessments.

#### 4.1.4 *The hub and spoke model evolving as a movement*

“Experiences of Information Village Research Project (IVRP), which was initiated in the year of 1998 in Pondicherry led to the launch of Mission 2007 which aims to forge multi-stakeholder partnerships for replicating the IVRP model in 600,000 villages by 15th August of 2007, India’s 60th independence day” [26]. Two years after the launch of Mission 2007: Every Village a Knowledge Centre, and the recommendation of the National Commission on Farmers to establish VKCs to bridge the urban–rural digital divide, there has been development at different levels to establish 100,000 ICT-based common service centres, leveraging the State Wide Area Network (SWAN) infrastructure to provide internet connection to all the villages, setting up VRCs at the block level by the Indian Space Research Organization (ISRO) [27].

The innovative application of the Hub and Spoke Model in India became the foundation for Mission 2007 with a simple but powerful concept (every village a knowledge centre) underlying it. The mission has successfully offered an ecosystem for Village Knowledge Centres (commonly referred to as telecentres) that are presently meant to bridge the various divides that exist in India. The Mission has ensured commitments from multi-stakeholders for establishing telecentres in each of the 240,000 *Panchayats* in India. The Mission has brought together a coalition of the concerned that is engaged in playing appropriate roles. The Mission has now become a movement, which the nation is willing to sustain for the next generation. The rural knowledge movement—*Grameen Gyan Abhiyan*—is an invention of Mission 2007. The idea of 600,000 has now become a government policy with the announcement made

by the then Hon'ble President in June 2009. India will have 240,000 Bharat Nirman Community Knowledge Centres by 2012 [28]. Given the potential and ubiquitous nature of technology, multiple stakeholders began to extend ICT-based knowledge for reaching more communities across India.

## 4.2 Results from the assessment

During the assessment period (April 2017 to March 2018), 1635 farmers including 506 women had accessed the knowledge hubs seeking real time information and knowledge exclusively on agriculture in 7 villages from Thanjavur and Pudukkottai districts in Tamil Nadu. Even though diverse tools have been used to disseminate information to the farming community, for the benefit of the study, three tools namely VKC, Voice SMS and Helpline that are connected with the hub of VRC for its functionality were taken to understand the benefits of the farmers. The data shows that all the three tools have provided required information and knowledge to address the issues related to farm operations.

### 4.2.1 Access to knowledge through ICT among farming community

The qualitative data indicates that the farmers were deprived of information concerning farming practices prior to the establishment of VKCs. The introduction of hub and spoke model of VRC and VKC ensured equity across caste, class, and gender in the flow of information to the community. Men and women have appreciated the access to information and need-based knowledge provided for diverse areas including cattle management, education, employment, besides agriculture. The data gathered from the 300 respondents, covering 39% of women and 61% of men in the community, revealed that all of them have accessed information through multiple ICTs on a regular basis from the VRCs and VKCs.

It is inferred that the farmers have approached the VRCs, VKCs and accessed its ICT action platforms such as voice messages and helpline facility for addressing farm related issues for different crops such as paddy, vegetables, oil seeds, flowers, pulses, and sugarcane. They contacted VKCs primarily for addressing the issues pertaining to pest and disease management, nutrition deficiency, soil health, and seed treatment besides information on government schemes and market prices. It is inferred that the VRC extended support as required by the VKC other than sending voice messages and handling helpline calls.

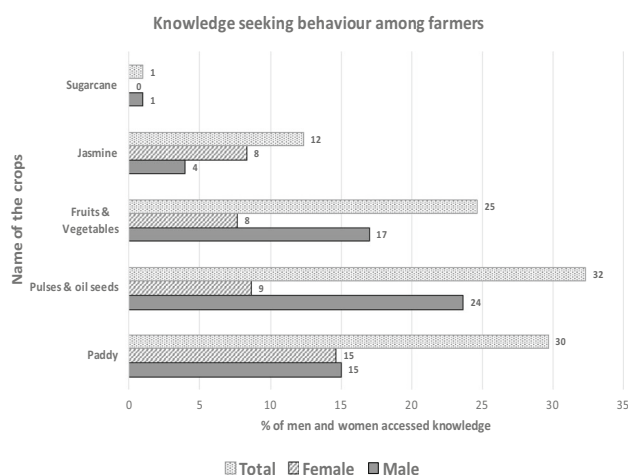
### 4.2.2 Gender participation: crop advisories through different ICTs

The data highlights that the hub and spoke model offers opportunity for both men and women farmers to access information through diversified ICTs. Women are boldly accessing new spaces that give them the freedom to step out from the four walls of their homes to participate, make valuable and meaningful choices, be informed and apply new and key information via VKC and other ICTs, for bettering their knowledge.

### 4.2.3 Knowledge-seeking behaviour among men and women farmers

Figure 4 indicates the significance of the knowledge needs among both men and women farmers in agriculture. They accessed different ICT models to address their information and knowledge needs on best cultivation practices, seed treatment, irrigation techniques, market price, pest and disease, government subsidies, nutrition management related to pulses and oil seeds, paddy, fruits and vegetables, jasmine, and sugarcane. There are similarities and differences in the knowledge-seeking behaviour of men and women for addressing the issues in different crops, which also resonate and contrast with the existing gender division of labour.

The access to information and knowledge for paddy cultivation among men and women indicates a balance (15% each). The percentage of knowledge accessed by male farmers on pulses and oil seeds (24%), and fruits and vegetables (17%) is high compared to women (9 and 8% respectively). At the same time, men's involvement in accessing knowledge on fruits and vegetable cultivation emphasizes that there is a shift from the stereotypical role



**Fig. 4** Type of information and knowledge accessed by men and women farmers for different crops

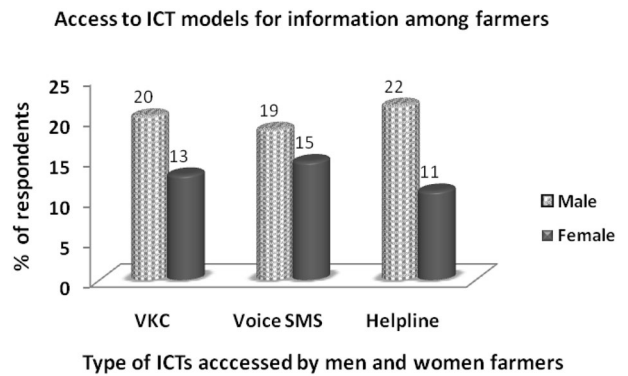


of men in agriculture. However, the greater information and knowledge needs among women for floriculture reiterates the gender stereotype of women in agriculture. Male farmers alone accessed the information on sugarcane, and it is inferred that it is they who are predominantly involved in the operations given its status of contract farming.

Over time there has been a change in the type of knowledge that both men and women farmers sought through ICT. From the qualitative data and trend of knowledge services rendered through VRC and VKC, it is inferred that in the past, women were keen to access information and knowledge related to livestock rearing, floriculture, and horticulture besides health, water, and enterprise development. Similarly men accessed knowledge related to pest, disease, fertilizer management, market information and government entitlements. This study revealed that the knowledge needs on best cultivation practices, techniques of seed treatments, pest and disease management, soil health and integrated nutrient management, market information, and latest technologies in agriculture are accessed by both men and women farmers. The shift in the knowledge sought by men and women farmers points to a shift in gender roles in agriculture. It underscores the growing feminisation of agriculture, i.e., the growing diversification of roles played by women in agriculture which goes far beyond their traditionally designated work to other strategic interventions in agriculture. The availability of hub and spoke model of VRC and VKC with its other ICT platforms is thus valued by them as they cater to their information- and knowledge-needs.

#### 4.2.4 Access to different technology among farming community

The data shows that with the hub and spoke model, the farming communities who had no prior access to ICTs and scientific content specific for increasing crop yields, gained access to ICTs and need-based information. The data in Fig. 5 indicates that in terms of seeking advisories, all 3 tools—VKC, voice SMS and agriculture helpline (33, 34, and 33% respectively) were equally useful. However, the data indicates that both men and women accessed information through different ICTs. Overall, men have outnumbered women by 20% in accessing information technologies for agriculture-related advisories. Of the 39% of women who accessed technology, it is inferred that 15% of them have accessed information through mobile-based voice messages, 13% through VKCs, and 11% through the agriculture helpline. In contrast, the preference of tools for accessing information among men farmers are helpline (22%), VKC (20%) and voice SMS (19%). From the qualitative data, it is inferred that tailoring of content to meet the needs of women and disseminating it at their



**Fig. 5** Access to different ICTs by men and women farmers for addressing knowledge needs

convenience played a critical role in ensuring their access to voice messages. Most women farmers indicated that they are able to access the advisories at their preferred time when their spouse or sons are around with their mobile handsets. Men preferred helpline as a means for accessing information, as they can directly discuss with the experts using their handsets. VKC being available at their village, both men and women have accessed it for information.

A study carried out by MSSRF captured a story of Manoharan Annavasal Sadasivam of Thiruvaiyaru, a 52-year-old farmer. He said “I wanted to learn more about agriculture and farming. It has been very easy to obtain knowledge and information on agriculture through the Village Knowledge Centre. For cultivating banana, I followed the method suggested in the knowledge portal *Valam*” (Resource). [This is a predefined locale specific knowledge package on cultivation and management practices of different crops.] “I cultivated four acres of land with banana crop, with the local banana variety *adukku monthan*. Before the planting of the stems, as advised through the Knowledge Centre, I treated them with the Bavistin mixture and this prevented root rot. Earlier it would cost me about Rs 6000/-per acre to cultivate, but now the cost has come down to Rs 4000/-per acre. I was able to get a profitable yield of Rs 12,000/- to Rs 15,000/-per annum. I had applied fresh compost once in 3 months, and alternated with phosphate, potash, and urea thrice. With this, I was able to obtain a good harvest. Whenever I have any further doubts on agriculture or my crops, I am happy that I can approach the Village Knowledge Centre to learn more” [15].

#### 4.2.5 Pattern of ICT usage for crop advisories

Table 2 affirms that there is a difference in the access ratio of men and women to different ICTs for accessing agro advisories for different crops such as paddy, pulses and oil seeds, fruits and vegetables, jasmine, and sugarcane. Male

**Table 2** Men and women farmers who accessed agro advisories for diverse crops using different ICTs

| Crops                 | VKC  |        | Voice SMS |        | Helpline |        | All |
|-----------------------|------|--------|-----------|--------|----------|--------|-----|
|                       | Male | Female | Male      | Female | Male     | Female |     |
| Paddy                 | 15   | 14     | 15        | 16     | 15       | 14     | 89  |
| Pulses and oil seeds  | 20   | 02     | 23        | 14     | 28       | 10     | 97  |
| Fruits and vegetables | 22   | 08     | 11        | 06     | 18       | 09     | 74  |
| Jasmine               | 04   | 15     | 04        | 08     | 04       | 02     | 37  |
| Sugarcane             |      |        | 03        | 00     |          |        | 3   |
| All                   | 61   | 39     | 56        | 44     | 65       | 35     | 300 |

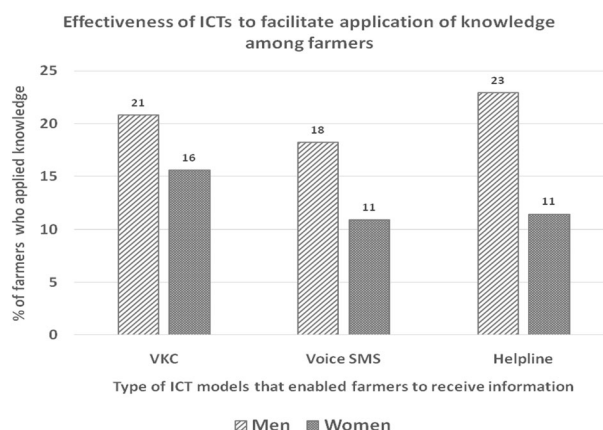
participation is higher in accessing ICTs to address the knowledge needs in relation to crops such as pulses and vegetables. More men farmers cultivate fruits and vegetable, which reiterates a shift in their roles. It is inferred that men farmers accessed all three tools equally for addressing their knowledge needs for paddy and jasmine crops. Their preference to access information related to fruits and vegetables are VKCs and helpline. The data shows that women farmers used voice message and helpline to address their knowledge needs on paddy, pulses, and oil seeds. The VKC helpline was their preferred choice to access information on fruits and vegetables, and for jasmine, it was the VKC.

The qualitative data collected from the farmers' states that their satisfaction with the solutions obtained from the ICT platforms to crop-related issues is high, as the advisories rendered are timely, need-based, locally-relevant and affordable in terms of cost. The process undertaken at the VRC, VKC, and helpline follow a two-way communication to identify the problem in the crop prior to the recommendation. The respondents also stated that the voice messages help to get advisories and form a base to contact either VRC or VKC or helpline to get further information as required.

The data signifies that dissemination of need-based knowledge that are applicable to the local context, its cost effectiveness and result orientation using two-way communication are in demand among farmers.

#### 4.2.6 Application of the knowledge offered through hub and spokes model

Of all the farmers who accessed farming-related information and knowledge, 64% of the farmers applied the knowledge received through ICT enabled hub and spoke model. Of the farmers who accessed farming-related information 62% were men and 38% women. The data in Fig. 6 indicates that 37% of farmers (21% men and 16% women) could apply the knowledge received through VKCs, while 34% of farmers (23% men and 11% women) applied the recommendations offered by the helpline and

**Fig. 6** Effectiveness of hub and spoke model as a pathway for addressing Knowledge needs of farmers

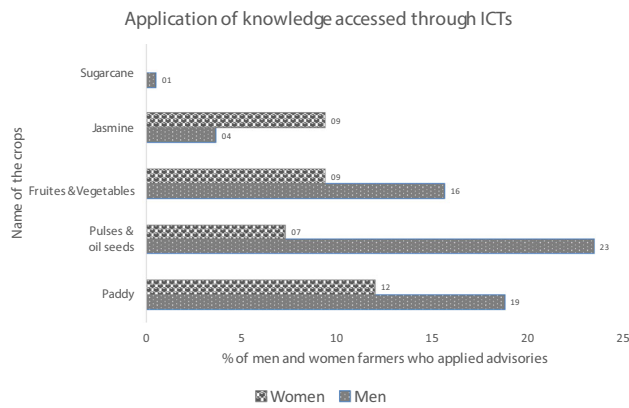
29% of farmers (18% men and 11% women) based on the recommendations through voice SMS.

The application of agro advisories realized through VKC (37%) followed by helpline (34%) indicates that the opportunities of two way communication adds value and enables men and women farmers to clarify and apply the knowledge better. Between VKC and helpline, VKC was found to be an effective tool as it is located at their village with an ample knowledge base to facilitate access to immediate knowledge for the farmers either from VKC or connect to appropriate knowledge holders through VRCs.

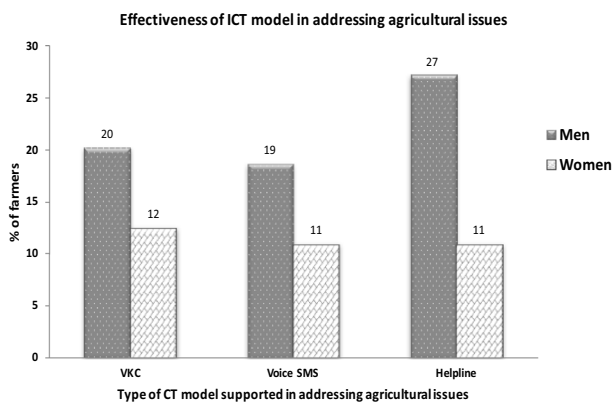
The application of knowledge by men farmers is found in crops such as pulses and oil seeds (23%), paddy (19%), fruits and vegetables (16%), jasmine (9%) and sugarcane (1%) as given in Fig. 7. Women farmers applied the knowledge accessed through hub and spoke model, particularly through VKC and helpline, for crops such as paddy (12%), fruits and vegetables and jasmine (9% each), and pulses and oil seeds (7%).

#### 4.2.7 Effectiveness of hub and spoke model in providing real time solutions

As presented in Fig. 8, of the farmers who applied the knowledge received from the hub and spoke model, 67%



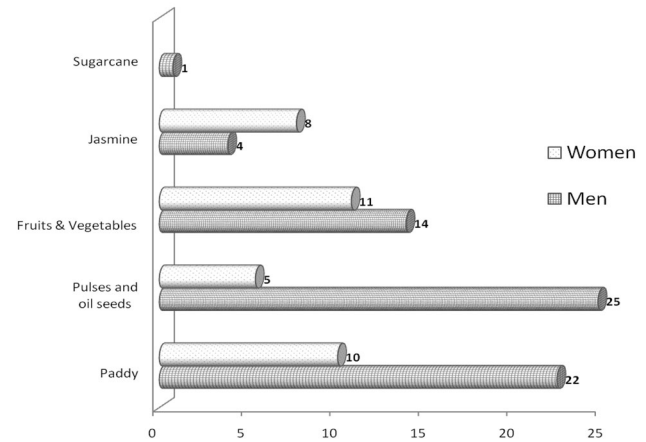
**Fig. 7** Application of knowledge by men and women farmers in different crops



**Fig. 8** Role of hub and spokes model in addressing crop issues

stated that they were able to solve the issues in their crops by applying the recommendation. Among them 66% were men and the remaining 34% were women farmers. The data indicates that both VKC and helpline played an important role in providing actionable knowledge for informed decision making and in addressing the issues with the support of VRCs. Both the tools provided opportunities for farmers to interact with experts. The farmers have also indicated that the KWs of their own villages are acting as conduits to facilitate knowledge transaction between farmers and experts, which ensure community ownership and ensures sustainability. The KWs also facilitated the connection of farmers with the helpline, thus allowing them to have detailed discussions with the experts. This was made possible as VKC accommodates diverse tools to leverage knowledge resources to support farmers in need. Helpline services for agriculture that are being supported by VRC are recognized as successful tools by the farmers as they helped them acquire the required knowledge to address crop-related issues especially with relation to pest and disease management.

**Role of ICT in addressing issues in different crops**



**Fig. 9** Role of hub and spokes model in addressing different crop related issues

Figure 9 reveals that the men farmers were able to address the problems in cultivation of pulses and oil seeds by 25%, paddy by 22%, fruits and vegetables by 14%, jasmine by 4% and sugarcane by 1%. Women were able to address their crop-related issues for fruits and vegetables by 11%, paddy by 10% and floriculture by 8%. From the data it is inferred that of 43% of farmers, 70% of them could address their issues by means of VKC and helpline services. It reinforces the significant role of VRC and VKCs in agriculture in strengthening the information network with scientific experts. It is also indicative that compared to one way communication through voice messages, the other hub and spoke models such as VKC and helpline that engaged farming community through discussions brought forth better absorption and application of the knowledge and could realize results.

#### 4.2.8 Results from short case studies

The case studies are empirical evidence from some of the respondents in Tamil Nadu, which encapsulate the role of VKCs as spokes for providing timely access to relevant information. This was then instrumental in bringing an effective shift in farming practices that enhanced the yields among small holding men and women farmers. The qualitative data captured among the farmers indicates a shift in the pattern in the type of knowledge and approach of knowledge delivery, which corresponds well with the capacity advancement of the farmers, VKC, and KW in keeping with technology growth.

Ms. Jamuna lives with her husband and children in Paavasamy Agraharam in Thiruvaiyaru village and grew jackfruit in their backyard. She was happy with the yield and the income as it required very minimum attention, until a point when the branches of the tree began to wilt and

there was a sudden decrease in the jackfruit yield. Ms. Jamuna sought help from the Knowledge Worker, who had suggested spraying Bordeaux mixture in a given measure on the tree. Keeping with their recommendation, the soil was tilled, the wilted branches removed, and Bordeaux paste was applied where required. After 3 days, the Bordeaux paste was applied at 50 g mixed with 500 g of sand. In addition, 10 g of Bordeaux mixture was dissolved in one litre of water and used to thoroughly drench the bottom of the tree. This treatment was helpful in reviving the tree and it has now started bearing fruits and the yield has increased.

Mr. M. Murthy and Mr. T. Gurumurthy a small land-holding farmer and a daily wage labourer from Thirupalanam village of Thanjavur district had difficulty in accessing information that would improve the yield of paddy. Through the KW, Mr. Mohan of the VKC, both of them learnt to increase the yield by using *Pseudomonas*, *Azospirillum* and phosphobacterium for the seed treatment which will prevent root rot, blight etc. Mr. M. Murthy had learnt more about the application and proportionate use of fertilizers and seed treatment to increase the yield. He said, “earlier for 100 *kuzhi* (33 acre), I used to get 8 bags, now there is an increase of 3 to 4 bags of paddy”. Thanks to the scientific advisories received from the VKC, he began to apply the intensive farming practice of growing lentils (black gram) and soya, after the harvest of paddy. Likewise, Mr. T. Gurumurthy was cultivating paddy and sugarcane and was getting a fair income from his small holdings. The advisories from the VKC on the method of seed treatments and the choice of fertilizers brought gains for Mr. Gurumurthy, wherein he has been able to reduce his expenses by INR 1500/-per acre from INR 6000/-per acre and his profit has increased annually. Moreover, after harvesting paddy, both these farmers have adopted an intensive farming practice of growing crops like sunflower, pumpkin, ash gourd, sesame. This entire cropping system has incurred less input cost and increased their profits.

In the village of Merpanaikadu, of Pudukkottai district, Mr. M. Manickam cultivated paddy in one acre of land. A senior citizen, he has been using “Adipattam Saraswathi” paddy variety. Every time the crop would wilt and turn pale yellow after 40 days. He relied extensively on the pesticides and chemicals from input dealers to treat the issue and yet the issue persisted, in spite of repeated attempts. He had heard much about the VKC and its role in supporting the farmers in improving their yields, leading him to seek support from the KWs. Mr. Manickam, had taken the wilted samples to the VKC where he was first informed of the infestation as rice tungro virus which is transmitted through insects and that current pesticides would not control the disease. The advisories given were to follow the guidelines of the leaf colour chart and follow a soil

application of 2 kg *Trichoderma viride*, a bio fungicide and spraying of 500 ml of chlorpyrifos to control the vector. This helped resolve the infestation.

A study conducted by MSSRF brought out that the information regarding government schemes and subsidies reach people very fast now. The residents are able to get the benefits of schemes/subsidies. Mukundan (Agalankam village of Nagapattinam district) was happy to mention that four entitled farmers have got drillers with 50 per cent subsidy (from the agriculture department) which is likely to reduce expenses of other farmers in the village. He also mentioned a few other examples of eligible households getting subsidies on tarpaulin and chicken (through the BDO). He mentioned that nearly 72 households could utilise the subsidy given for buying chicken. It is mainly because they have been able to get the information on time [29].

## 5 Conclusion

The hub and spoke model of VRC and VKC is an innovation demonstrated by M S Swaminathan Research Foundation for sustainable rural development and knowledge empowerment using ICTs. This model has enabled VKC to act as a hub on its own over a period of time and has given the opportunity of expanding the partnership base with the knowledge holders. It also signifies the value that the community placed on the VKC model as a critical asset for meeting their knowledge needs and to be able to advance it with a sense of ownership to continue its knowledge services. The experience of different segments of the community clearly indicates that they have realized tangible and intangible benefits through VRC and VKCs. The tools like helpline have also been successful. However these types of independent tools are not self-sustainable as they require continued support from institutions or external members for regular functions, compared to the VKC model. Despite advances in technologies to provide information and knowledge, the findings unveil that locale-specific, scientific, demand-driven knowledge packages provided on time through user-friendly technology with the blend of adequate communication between the women and men farmers and relevant experts, alone can bring about a change. The findings also capture the need for leveraging the continuous feedback system to strengthen the delivery of need-based information, which remain dynamic even today, on a on-going basis for the farmers. The findings show that this model has also bridged the digital and gender gaps in the rural areas, particularly, have enabled the rural women to gain several intangible (and tangible) benefits. Additionally, the bottom up approach with a pro-nature, pro-poor, pro-women orientation in fulfilling the



real time knowledge needs of farming community becomes an important factor for the continuity of the hub and spoke model in enhancing knowledge capacities. Technologies will advance with time and the hub and spoke model might seem to become obsolete, but the horizon it has opened to link information seekers and information holders in a time- and locale-specific manner remains unchanged. The study also captures that a ICT-enabled model specifically for strengthening access to timely information to build the resilience capacity of the community, would strengthen their livelihoods in the context of climate change. Moreover, this model is adaptable and can be applied across communities for building sustainable cities and communities.

## References

1. Babu SC, Joshi PK, Glendenning CJ, Asenso-Okyere K, Rasheed Sulaiman V (2013) The state of agricultural extension reforms in India: strategic priorities and policy options. *Agric Econ Res Rev* 26(2):159–172
2. ICAR (1998) National agricultural technology project, man document. Indian Council of Agricultural Research, New Delhi
3. Rasheed Sulaiman V (2003) Agriculture extension: involvement of private sector. Occasional paper 29, Department of Economic Analysis and Research. National Bank for Agriculture and Rural Development, Mumbai. <https://www.nabard.org/demo/auth/wri/tereaddata/File/OC%2029.pdf>
4. Kesavan PC (2017) Swaminathan: legend in science and beyond. World Scientific Publishing, Singapore, p 260
5. Mishra DC (1996) Agricultural extension effectiveness in India: a cross sectional analysis. *Indian J Ext Educ* 32:1–4
6. Mukherjee A, Maity A (2015) Public-private partnership for convergence of extension services in Indian agriculture. *Curr Sci* 109(9):1557–1563
7. Planning Commission (2001, 2005, 2006, 2008) Approach paper to the tenth five year plan (2002–2007), midterm appraisal of the 10th five year plan (2002–2007), towards faster and more inclusive growth: an approach to the 11th five year plan (2007–2012) and eleventh five year plan (2007–2012). Government of India, New Delhi, vols I, III
8. Rasheed SV (2012) Agricultural extension in India: current status and ways forward. Background paper prepared for the roundtable consultation on agricultural extension, Beijing, March 15–17. Centre for Research on Innovation and Science Policy (CRISP), Hyderabad, India
9. M. S. Swaminathan Research Foundation (1991, 1992, 1994, 1998, 1999, 2000, 2002, 2004) Annual reports. [www.mssrf.org](http://www.mssrf.org)
10. Mittal S (2012) Modern ICT for agricultural development and risk management in smallholder agriculture in India. Socioeconomics working paper 3. CIMMYT, Mexico
11. National Sample Survey Organization (NSSO) (2005) Situation assessment survey of farmers: access to modern technology for farming, 59th round (January–December 2003), Ministry of Statistics and Programme Implementation, Government of India, New Delhi
12. Kesavan PC, Senthilkumaran S (2011) Rural development: livelihood with ecological integrity. Evergreen revolution—a blueprint for sustainable rural livelihoods and food security, focuses on achieving productivity without ecological harm while harnessing appropriate technologies. <https://www.geospatialworld.net/article/rural-development-livelihood-with-ecological-integrity>
13. Swaminathan MS (ed) (1993) Reaching the unreached: information technology a dialogue. Macmillan India Limited, Bangalore
14. Chambers R (1989) Land-to-lab, and land-to-land concepts and possible approaches. Paper presented in workshop-cum-seminar on soil and water conservation measures in dry land areas (red soils), Central Research Institute for Dryland Agriculture, Hyderabad
15. Vedavalli L (2009) Human dimensions of ICT in Rural India (case studies from Annavasal, Sempatti, Thangachimadam and Thiruvaiyaru—Tamil Nadu)—reflections by rural women and men
16. Bhattacharji B (1998) Enhancing citizenship skills in an information age. In: AMIC 7th annual conference: Asia's information marketplace, race for technology, content and competence, bangkok. Asian Media Information and Communication Centre, Singapore, 21–23 May. [https://dr.ntu.edu.sg/bitstream/handle/10220/2070/AMIC\\_1998\\_MAY2123\(VOL1\)\\_09.pdf?sequence=1](https://dr.ntu.edu.sg/bitstream/handle/10220/2070/AMIC_1998_MAY2123(VOL1)_09.pdf?sequence=1)
17. Julian G, Mohan P (eds) (2011). Strengthening resilience in post-disaster situations stories, experience and lessons from South Asia International Development Research Centre, Ottawa, Canada and Academic Foundation, New Delhi
18. Dixit S, Balaji V (2007) ICTs for 'second green revolution' in rainfed agro-ecosystem: issues and strategies. *Int J Ext Educ* 3:80–88
19. Vaidyanathan V (2008) Lessons from good telecentres. IT for change. <http://indiatgether.org/articles/telectr-government-2/print>
20. Swaminathan MS (2010) Science and sustainable food security: selected papers of M. S. Swaminathan. [https://books.google.co.in/books?id=XdPFCgAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=XdPFCgAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
21. Ramji M (2006) Information and communication technology and economic development: some observations on the socially excluded from Village Knowledge Centres in Pondicherry, India
22. Alberts Bruce (1999) Presidential address from the science and the world's future. US National Academy of Science, Washington
23. Subramanian R, Shaver L (eds) (2011) Access to knowledge in India: new research on intellectual property. Innovation and Development, Bloomsbury Academy. [www.bloomsburyacademy.com](http://www.bloomsburyacademy.com)
24. Nanda S, Arunachalam S (2010) Transformative impact of ICT—change stories from rural India. In: MSSRF book by Arundhati. <http://www.suchit.net/books/Transformative-Impact-of-ICT.pdf>
25. Lele U, Gandhi K (2009) M. S. Swaminathan Research Foundation at 21: report of the independent program review: 313. <http://59.160.153.185/library/node/589>
26. Atanu G, Shadrach B (2006) Taking ICT to every Indian Village: opportunities and challenges. *One World South Asia*. <https://ssrn.com/abstract=897910>
27. Swaminathan MS (2006) Every village a knowledge center. In: Kapila R, Kapila U (eds) India's economy: a journey in time and space. Academic Foundation, New Delhi, pp 259–262
28. Shadrach B, Senthil Kumaran P (2009) Mission to Movement, telecentre.org and Grameen Gyan Abhiyan Secretariat
29. Vedavalli L (2010) Integrated village development through knowledge empowerment: the experience of Agalankannur village. MSSRF, Chennai